

Ask

Statistics is a field that deals with collection, organization, analysis, interpretation & presentation of the data

↓
Decision Making

Eg: Age = {24, 27, 14, 13, 28, 29, 31, 32}

↓
{Online shopping}

(i) Mean

(ii) Median

(iii) Distribution

charts & graphs

(i) Histogram

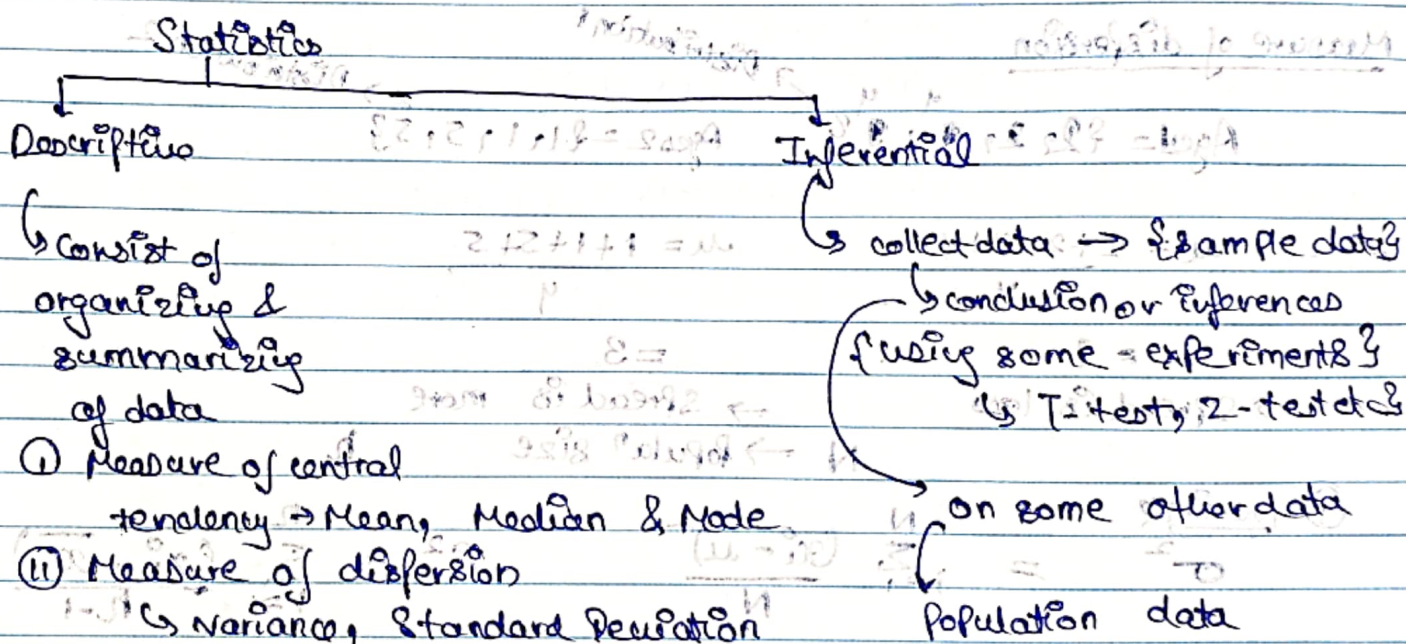
(ii) Pdf

(iii) Cdf

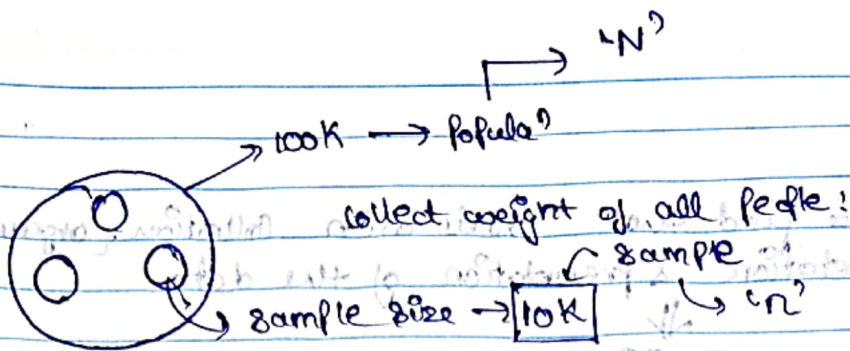
→ understand the data and make some decision → to grow your business

Applications

- ↳ (i) Everyone uses it
- (ii) Machine learning / Data Science
- (iii) Data analyst
- (iv) Business Analyst



Popular & sample data



Measure of central Tendency

Mean (μ) = $\frac{\sum_{i=1}^N x_i}{N}$ $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$

~~Median~~

Median

way of first sort values → then central element will be median

Mode

we select the element which has max freq

Measure of dispersion

Age1 = {2, 2, 4, 4, 4} Age2 = {1, 1, 5, 5}

Distribution 1 Distribution 2

$$\mu = \frac{1+1+5+5}{4}$$

$$= 3$$

→ Spread is more

N → Populaⁿ size

① Variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$\mu = 3$$

$$\text{Age1} =$$

x_i	u	$(x_i - u)^2$
2	3	1
2	3	1
4	3	1
4	3	1
		<u>4</u>

$$N=4$$

$$\Rightarrow 4/4 = 1$$

variance

x_i	u	$(x_i - u)^2$
1	3	4
3	3	0
5	3	4
5	3	4
		<u>16</u>

$$N=4$$

$$\Rightarrow 16/4 = 4$$

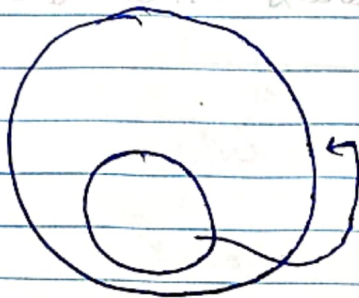
Sample variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Bessel correction

why $(n-1)$?

Imp



Inferences

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

underestimating the true population variance

Ages of kids ... 2, 2, 2 ... huge ... $\bar{x} \approx \mu$... $s^2 \approx \sigma^2$

Sometimes $\bar{x} \ll \mu$... $s^2 \ll \sigma^2$

Populations

Population (N)

Sample (n)

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

$$\text{std} = \sqrt{\sigma^2}$$

$$\text{Std} = \sqrt{s^2}$$

How far you are away from mean

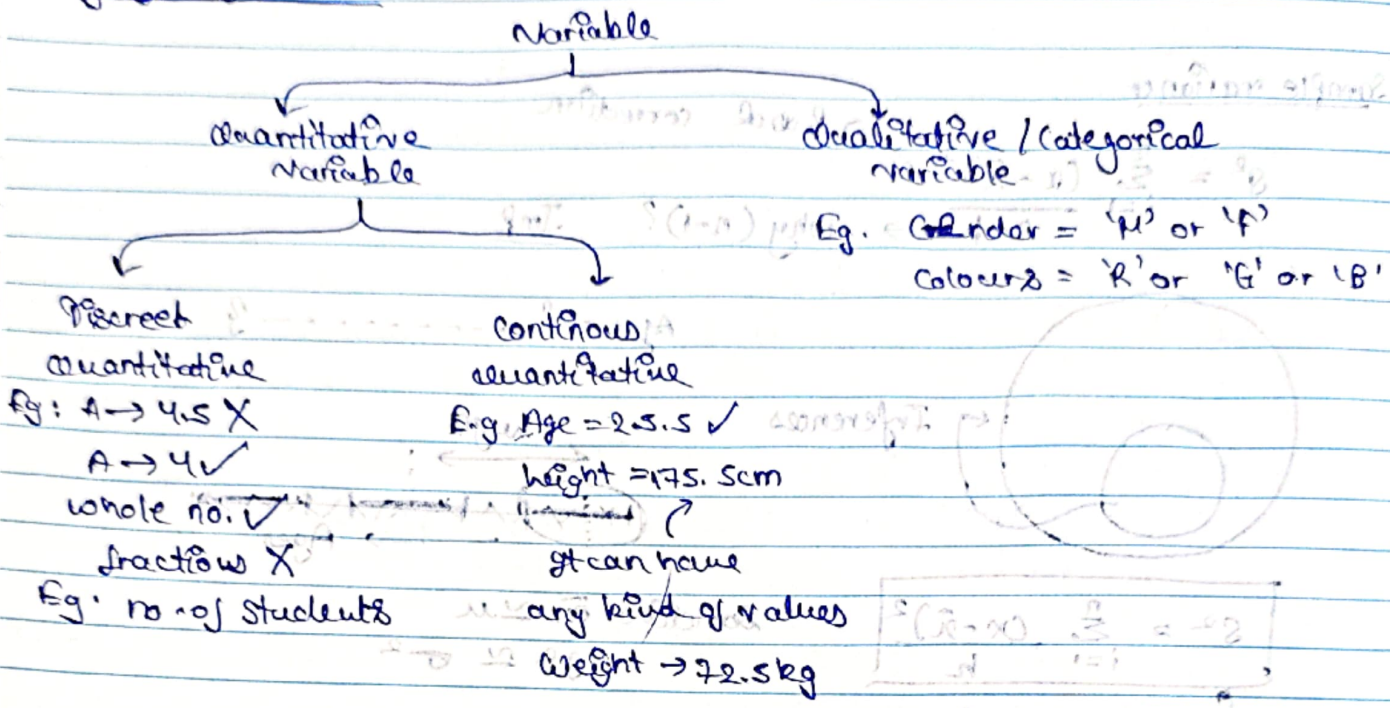
Variables

Defn: Property that can take up any value

Age = 25 ✓ Gender = 'M' ✓

Ages = [25, 26] X Height = 7.2 ✓

Dif. Types of variables



Random Variables

$X \rightarrow$ function whose values are derived from Process or Exp.

$$X = \begin{cases} 0 & H \\ 1 & T \end{cases}$$

tossing a coin

$$X = \begin{cases} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{cases}$$

Rolling a fair dice

Random Variables

Discrete

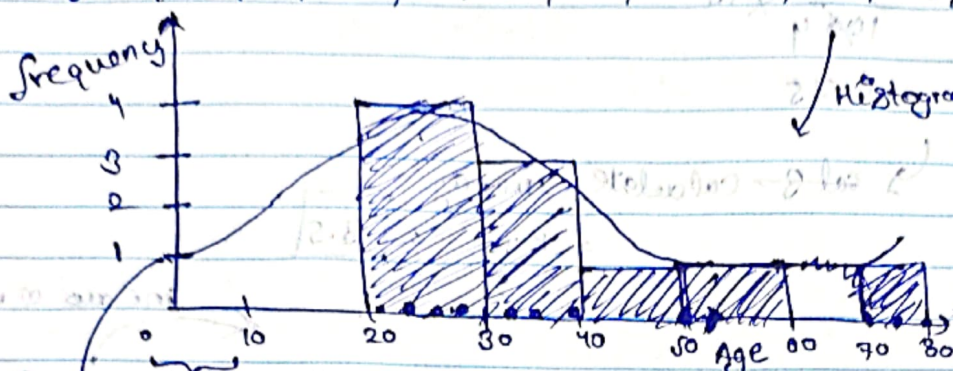
eg Tossing a coin
Rolling a dice

Random Continuous

eg Tomorrow how many inches it is going to rain

Histograms

$X = \{28, 24, 25, 30, 34, 36, 40, 50, 60, 75, 80\}$



Histogram 20 → 30: 4
30 → 40: 3
40 → 50: 1
50 → 60: 1
60 → 70: 0
70 → 80: 1

bins = 10
frequency of distribution among bins
smoother the histogram

{Kernel density estimation}

Percentiles & Quartiles

Percentage → {1, 2, 3, 4, 5, 6}

No. of odd = 3

percentage of odd no. in this group = $\frac{3}{6} \times 100 = 50\%$

Percentile: A percentile is a value below which a certain percentage of observation lies.

{2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 8, 9, 10}

Percentile

of value $x = \frac{\text{\# of values below } x}{n} \times 100$

$5 = \frac{11}{20} \times 100$

$\rightarrow$ It means that 78.57% of entire distribution is below 9
 $\approx 78.57\%$ of value 9
 Percentile Ranking

$$\rightarrow \text{Value} = \frac{\text{Percentile} \times n+1}{100}$$

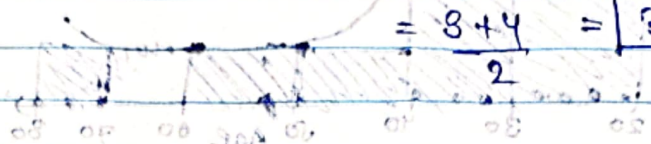
$$= \frac{25}{100} \times 14+1$$

$$= \frac{251}{1004} \times 15$$

$\rightarrow$ calc $\rightarrow$ calculate average

$$= \frac{8+4}{2} = 3.5$$

Quantiles

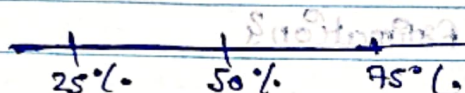


25% = 1st quartile

50% = 2nd "

75% = 3rd "

$\rightarrow$ Percentile



5 Numbers Summary

(i) minimum

(ii) 1st quartile (25%) 1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27

(iii) Median 12 = 12

(iv) 3rd quartile (75%)

(v) Maximum

Lower fence value, higher fence value
 [Lower fence value, higher fence value]

$$\text{Lower fence} = Q_1 - 1.5(IQR)$$

$$\text{Higher fence} = Q_3 + 1.5(IQR)$$

$Q_1 \rightarrow 25\%$

$$\rightarrow \frac{25}{100} \times (n+1)$$

$$= \frac{25}{100} \times 28 = 5^{th} \text{ position} = 3$$

$$Q_3 = \frac{75}{100} \times 20$$

$$= 15^{\text{th}} \text{ pos}^n$$

$$= 7 + (15 - 14) \times \frac{8 - 7}{15 - 14}$$

$$IQR = Q_3 - Q_1$$

$$= 7 - 3$$

$$= 4$$

$$\text{lower fence} = Q_1 - 1.5(IQR)$$

$$= 3 - 1.5(4)$$

$$\text{higher fence} = Q_3 + 1.5(IQR)$$

$$= 7 + 1.5(4)$$

$$= 13$$

$$\text{range} = [-3, 13]$$

~~outlier~~ $\rightarrow$ outlier

min: 1

max: 9

So, min: 1, Q_1 : 3, median: 5, Q_3 : 7, max: 9

5 no. summary

can help define box plot

Covariance & Correlation

x y random variable

x y

2 3

4 5

6 7

8 9

[Quantify the relation b/w x & y]

if $x \uparrow$ $y \uparrow$

or

$x \downarrow$ $y \downarrow$

or

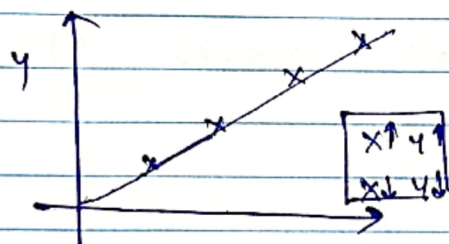
$x \uparrow$ $y \downarrow$

or

$x \downarrow$ $y \uparrow$

$\Rightarrow$ covariance

OF =



$x \uparrow$ $y \downarrow$

$\Rightarrow$ +ve Covariance

$$(0.5 - 0.5)(3 - 5) + (0.5 - 0.5)(5 - 7) + (0.5 - 0.5)(7 - 9) =$$

$$(0.5 - 0.5)(5 - 7) + (0.5 - 0.5)(7 - 9) + (0.5 - 0.5)(9 - 11) =$$

$$(0.5 - 0.5)(9 - 11) + (0.5 - 0.5)(11 - 13) + (0.5 - 0.5)(13 - 15) =$$

$$(0.5 - 0.5)(15 - 17) + (0.5 - 0.5)(17 - 19) + (0.5 - 0.5)(19 - 21) =$$

$$(0.5 - 0.5)(21 - 23) + (0.5 - 0.5)(23 - 25) + (0.5 - 0.5)(25 - 27) =$$

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$$(0.5 - 0.5)(579 - 581) + (0.5 - 0.5)(581 - 583) + (0.5 - 0.5)(583 - 585) =$$

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$$(0.5 - 0.5)(591 - 593) + (0.5 - 0.5)(593 - 595) + (0.5 - 0.5)(595 - 597) =$$

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$$(0.5 - 0.5)(609 - 611) + (0.5 - 0.5)(611 - 613) + (0.5 - 0.5)(613 - 615) =$$

$$(0.5 - 0.5)(615 - 61$$

Covariances

$$\text{Cov}(X, Y) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\text{Cov}(X, X) = \sum_{i=1}^n \frac{(x_i - \bar{x}) \cdot (x_i - \bar{x})}{n-1}$$

$$\boxed{\text{Cov}(X, X) = \text{Var}(X)}$$

$x_i \rightarrow$ data points of random variable x

$\bar{x} \rightarrow$ Sample mean of x

$y_i \rightarrow$ Data points of random variable y

$\bar{y} \rightarrow$ Sample mean of y

Std $\Rightarrow$

Hours Studied	Exam Scores
2	50
3	60
4	70
5	80
6	90

$x \uparrow y \uparrow$

$x \downarrow y \downarrow$

$\Rightarrow$ +ve Covariance

$$\bar{x} = \frac{3+2+4+5+6}{5}$$

$$= 4$$

$$\bar{y} = \frac{50+60+70+80+90}{5}$$

$$= 70$$

$$\text{Cov}(X, Y) = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$= (2-4)(50-70) + (3-4)(60-70) + (4-4)(70-70)$$

$$+ (5-4)(80-70) + (6-4)(90-70)$$

$$= (-2)(-20) + (-1)(-10) + 0 + 1(10) + 2(20)$$

47

$$\text{Cov}(X, Y) = 20$$

→ The +ve covariance indicates the no. of hours studied increases the exam score.

X	4	6
Y	10	12
	12	14

→ -ve covariance

✓ → quantify the reln b/w x and y

✗ → Covariance does not have a specific limit value
 $\text{Cov}(X, Y)$ can range up to $(-\infty \text{ to } +\infty)$

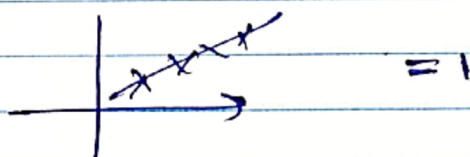
Correlation → Pearson Correlation Coefficient
 Spearman Rank

① Pearson → limits the value of corr. to $[-1, \text{to } 1]$

$$r_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

② The more the value towards the +1 → more +ve corr(X, Y)
 " " " " " " -1 → more -ve " "

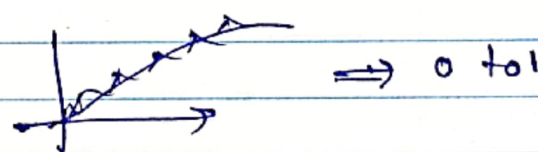
④ Spearman Rank correlation



Pearson correlation is

not able to capture non linear

pattern



Spearman = $\frac{\text{Cov}(R_x, R_y)}{\sigma(R_x) \times \sigma(R_y)}$

$\sigma = (y-x) \text{ var}$			
X	Y	Rank(X)	Rank(Y)
2	2	2	1
3	4	3	2
5	6	4	3
7	8	5	4
0	7	1	5

if there is a positive relationship between two variables, the correlation coefficient will be positive. If there is a negative relationship, the correlation coefficient will be negative. If there is no relationship, the correlation coefficient will be zero.

Correlation coefficient is a measure of the strength and direction of the relationship between two variables.

Correlation coefficient ranges from -1 to 1. A value of 1 indicates a perfect positive correlation, a value of -1 indicates a perfect negative correlation, and a value of 0 indicates no correlation.

$$\frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} = r$$

Correlation coefficient is a measure of the strength and direction of the relationship between two variables. It is calculated as the covariance of the two variables divided by the product of their standard deviations.

Correlation coefficient is a measure of the strength and direction of the relationship between two variables.

